

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/328797745>

Note: A versatile 3D-printed droplet-on-demand generator

Article in *Review of Scientific Instruments* · November 2018

DOI: 10.1063/1.5054400

CITATIONS

10

READS

821

2 authors, including:



[Daniel M. Harris](#)

Brown University

80 PUBLICATIONS 1,442 CITATIONS

[SEE PROFILE](#)

Note: A versatile 3D-printed droplet-on-demand generator

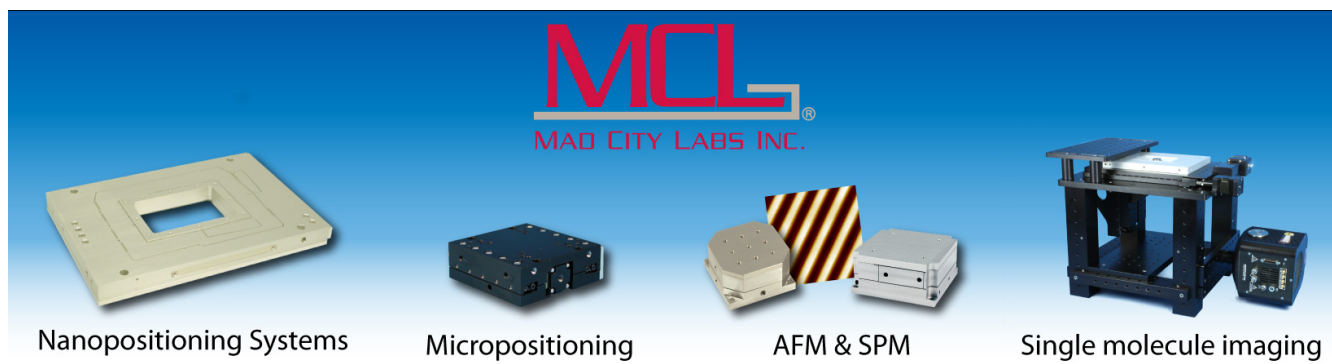
Nikolay Ionkin, and Daniel M. Harris

Citation: [Review of Scientific Instruments](#) **89**, 116103 (2018); doi: 10.1063/1.5054400

View online: <https://doi.org/10.1063/1.5054400>

View Table of Contents: <http://aip.scitation.org/toc/rsi/89/11>

Published by the [American Institute of Physics](#)



Note: A versatile 3D-printed droplet-on-demand generator

Nikolay Ionkin^{a)} and Daniel M. Harris^{b)}

School of Engineering, Brown University, Providence, Rhode Island 02912, USA

(Received 31 August 2018; accepted 18 October 2018; published online 7 November 2018)

A versatile 3D-printed droplet-on-demand generator is presented for laboratory use in droplet impact and similar experiments. The design described and tested in the present work is modeled off of an existing design [Harris *et al.*, *Exp. Fluids* **56**, 83 (2015)] but is tested with an extended range of working fluids, and the manufacturing process is greatly simplified by 3D-printing the principal components. The present device is tested with de-ionized water and water-glycerol mixtures and was reliably able to produce single droplets-on-demand of diameters 0.65–1.32 mm with an overall variability of less than 1%. The computer-aided design (CAD) files, parts list, sample software, and circuit layout are available with this note, allowing for the device to be readily reproduced or adapted for a wide range of experimental applications. *Published by AIP Publishing.* <https://doi.org/10.1063/1.5054400>

There is a rapid and persistent growth in the study of dynamical behavior of droplets; however, the precise generation of these droplets over a range of sizes can be challenging. One area of study where a reliable and precise droplet generator can be used is in studies of droplet impacts.^{2–5} The outcomes of such impact experiments are strongly dependent on the size of the drop created, and therefore experiments can be challenging to repeat without a reliable method for generating droplets.

Many designs for droplet generators have been proposed in the literature, but most remain complex and difficult to manufacture and operate for laboratory use. A simple solution that is often employed is gravitational detachment from a vertically oriented needle, although generating single sub-millimetric droplets with high-surface tension fluids (such as pure water) is impractical due to the minuscule scale of the needle that would be required for such operations. For reference, in a previous droplet impact study, a syringe with a fine needle of inner diameter 80 μm was used to produce pure water droplets of diameter 1.72 mm with an overall variability of $\pm 3\%$.⁶ Other generator designs have been proposed to achieve smaller drop sizes and are actuated by piezoelectric disks,^{7,8} loudspeakers,⁹ or compressed air¹⁰ but require machining equipment and/or significantly more complicated and expensive components which may include pressure regulators, sophisticated electronics, or custom nozzles. Despite the relative simplicity of our design from the perspectives of both manufacturing and operation, it exceeds such previous designs in terms of drop-to-drop and day-to-day drop size consistency. The design described and tested in the present work is based off a droplet generator previously developed.¹ This predecessor to the present design used a piezoelectric actuator driven by a square pulse and was shown to generate single droplets of 20 cSt silicone oil ranging from 0.5 to 1.4 mm in diameter with less than 1% variability.

The droplet generator described herein aims to further simplify fabrication by decreasing the amount of active time

needed to fabricate the components by having a majority of the components 3D printed. We have also tested the device with a wider range of fluids in order to demonstrate its versatility so that it may be readily used in a much wider range of experiments. In particular, the droplet generator is tested with de-ionized water and several glycerin-water solutions. The device maintains a drop-to-drop and day-to-day usage variability of less than 1%. Four primary components constitute the droplet generator which include a piezoelectric actuator, nozzle, fluid reservoir, and fluid chamber, as shown in Fig. 1. The piezoelectric actuator is placed at the top of the fluid chamber and clamped down using a 3D-printed ring and threaded bolts. An O-ring prevents air from entering the system as well as fluid from escaping. As a voltage is applied to the piezoelectric disk, the diaphragm flexes and the working fluid exits from the tip of the nozzle located below the fluid chamber.

The fluid chamber as well as other primary components is similar in dimensions to its predecessor.¹ The generator body itself consists of an inner cylindrical opening of 28.6 mm diameter and depth 9.7 mm which tapers down to a 3D-printed nozzle threaded onto the bottom and is sealed on top with a piezoelectric disk (35 mm diameter). In the present design, all components are 3D printed from a clear resin (RS-F2-GPCL-04 Formlabs) on the commercially available Formlabs Form 2 Stereolithography (SLA) printer. The clear resin allows for a quick method to determine whether or not air bubbles are present within the system, as well a method to visually determine the amount of fluid within the system. Six nozzles are printed with the outlet diameters ranging from 0.46 to 1.26 mm. The nozzles are fabricated with a Dental Model resin (RS-F2-DMBE-02 Formlabs) which is claimed to have an increased printing accuracy over the clear resin. It is found that nozzles printed with an outer diameter less than 0.46 mm tend to be less circular and have irregular edges. Furthermore, the diameter of the nozzles tends to differ from the designed value and thus is measured optically after being printed to determine an accurate outlet diameter. Forcing compressed air through the nozzle before the alcohol bath and final UV treatment is also essential to help retain a more accurate diameter.

Using a clear tube, the fluid chamber is connected to a 3D-printed fluid reservoir. Fluid within the reservoir remains at a

^{a)}Current address: Department of Mechanical Engineering, Columbia University, New York, NY, USA and Department of Physics, Clark University, Worcester, MA, USA.

^{b)}Author to whom correspondence should be addressed: daniel.harris3@brown.edu

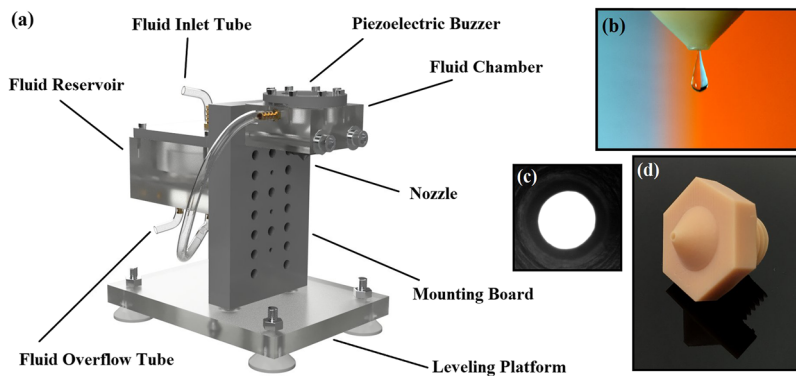


FIG. 1. (a) Rendering of the droplet generator using CAD software. Drawings, files, and parts list are available in the [supplementary material](#). (b) A single droplet of de-ionized water being formed at the nozzle against a blue and orange background. Image of the (c) outlet cross section and (d) body of a sample 3D-printed nozzle (diameter 1.02 mm).

constant level due to a concentric overflow chamber outside the reservoir. The fluid inlet tube at the top of the reservoir is mounted on a sink which acts as a de-ionized water source. Due to the fact that the sink was not constantly running, the reservoir was refilled for every 100 drops produced in our testing. Both the fluid reservoir and the fluid chamber are mounted on a 3D-printed vertical mounting board. The relative heights are chosen and fixed such that a static equilibrium exists between the meniscus at the nozzle and the fluid reservoir. The relative height in the present work is fixed at $\Delta h = 0$; however, as shown in prior work, a range of stable equilibrium heights exist $|\Delta h| \leq 2\sigma/\rho ga$, where σ and ρ are the surface tension and the density of the working fluid, g is the acceleration due to gravity, and a is the nozzle outlet radius.¹ It should be however noted that the droplet size depends on the value of Δh and so thus must be carefully fixed for repeatable performance.

The electrical components consist of a DC power supply (30 V), H-Bridge circuit, and Arduino Uno board. The power supply acts as the power source, and the digital pins on the Arduino Uno board are connected to the H-bridge circuit allowing for the voltage applied to the piezo to rapidly switch between +30 V and -30 V causing the piezoelectric disk to flex up and down. A positive voltage flexes the diaphragm down, forcing a droplet to form at the nozzle, and a negative voltage relieves the flex thereby retracting fluid at the nozzle outlet. The diameter of the nozzle and the pulse width for which the diaphragm is activated, the latter of which can be directly controlled by the Arduino Uno, determine the size of the droplet produced. The rate at which droplets are generated is controlled through the Arduino Uno by changing the time in between pulses. For this study, the drops were created at a period of 2 s. Further increasing the delay period showed no measurable difference in the droplet diameter or dynamics, indicating that we are operating in true droplet-on-demand mode in the present testing. However, we note that below a critical delay period for a given set of parameters there is a measurable influence on the droplet diameter and an even further decrease in the delay period can lead to irregular generation sequences where droplets are not produced at every pulse. Note that it is critical to turn off interrupts in the Arduino software during the pulse generation stage for accurate microsecond timing.

Before use, the system should be fully flushed with isopropyl alcohol to remove any contaminants. When using de-ionized water, it has been found that contaminants readily

affect the fluid properties and result in a change in the range of pulse widths over which droplets are generated. Following the alcohol flush, the system must then be flushed with de-ionized water to remove any residual alcohol. If glassware is used to fill the fluid reservoir, it too should thoroughly be cleaned to remove any contaminants. Then the system must be bled in order to remove any air bubbles. This can be performed by flipping the fluid chamber, lowering it well below the reservoir, filling the reservoir with the working fluid, allowing some fluid to flow from the inverted nozzle, covering the nozzle tip, and finally raising it back to the level of the reservoir. If any bubbles are present in the chamber, the piezoelectric disk will create a more audible clicking noise than when there are no air bubbles; however, air bubbles can typically also be seen through the clear resin. It is found that the presence of an air bubble shifts the range of pulse widths for which droplets are formed. Other precautions that should be taken are verifying the lack of air bubbles within any prepared working fluid, which is more common when working with higher viscosity fluids.

To determine the accuracy and reliability of the droplet generator, it is placed in between an LED light source (Godex SL60W) with a diffuser and a camera (Allied Vision Mako U-130B, 200 mm Nikon Micro Nikkor lens). The camera is triggered by the Arduino Uno board so that a single image is taken with a delay chosen between (10-70 ms) after a droplet is formed, with a delay selected such that the drop appears approximately spherical. To capture the dynamics of droplets as they leave the nozzle tip shown in Fig. 2, a high speed camera

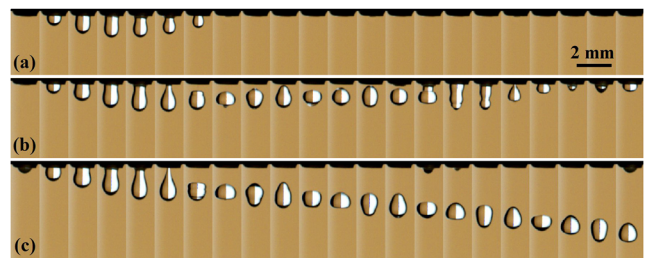


FIG. 2. Dependence of droplet generation for nozzle diameter 0.81 mm and applied voltage 30 V using de-ionized water. (a) No droplet is generated for pulse width 1100 μs . (b) No droplet is generated using pulse width 1225 μs . (c) A single 1.02 mm droplet is generated using pulse width 1500 μs . Neighboring images are taken 1.0 ms apart. A two-colored backdrop is chosen to increase droplet contrast.

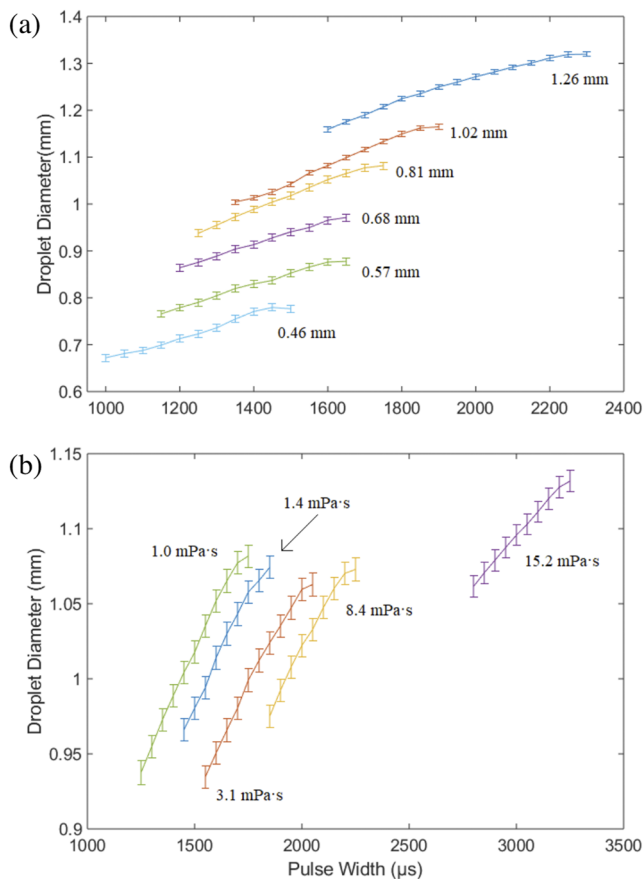


FIG. 3. (a) Droplet diameters for six different nozzles evaluated as a function of pulse width using de-ionized water with viscosity 1.0 mPa s. (b) Droplet diameters for nozzle diameter 0.81 mm as a function of pulse width for several fluid viscosities corresponding to mixtures of de-ionized water and glycerol. For both panels, a minimum of 30 droplets are measured for each data point.

(Phantom Miro LC311, 200 mm Nikon Micro Nikkor lens) is used.

The droplet diameters were determined from the images with a Canny edge detection algorithm in MATLAB,¹¹ and an equivalent spherical droplet diameter is calculated. The diameter of the droplet is then calculated in pixels and converted to mm by the ratio $5.10 \mu\text{m}/\text{pixel}$, which was determined with a microscope calibration slide. Error bars are included in all results to show pixel uncertainty along with actual variation in droplet sizes which is taken to be one standard deviation.

Each nozzle is tested to determine the range of pulse widths for which a single droplet is generated using pure de-ionized water. In Fig. 2, we highlight three possible regimes of pulse width using a 0.81 mm nozzle. The first regime is where the pulse width is too small for a droplet to form [Fig. 2(a)]. As the pulse width increases, a droplet is formed and separates but moves upwards initially and ultimately reconnects with the oscillating meniscus at the nozzle [Fig. 2(b)]. The final regime shown is when the droplet forms and falls away from the nozzle [Fig. 2(c)]. We note that as the pulse width is increased, the initial downward velocity of the detached droplet increases as well.

Figure 3(a) shows the range of pulse widths at which a droplet is formed for each of the six nozzles, ranging from a nozzle diameter of 0.46 to $1.26 \text{ mm} \pm 0.01 \text{ mm}$. Above

a certain pulse width, the regular increasing trend is broken and the droplet reaches a maximum diameter. Although the droplet diameter may continue to increase past this point, it does not do so in a consistent manner, often the primary droplet is accompanied by the appearance of undesirable satellite droplets.

To further explore the potential of the droplet generator, several different fluid viscosities were tested, as shown in Fig. 3(b). Various viscosities were achieved by thoroughly mixing a solution of de-ionized water and glycerin at an ambient temperature of $21^\circ\text{C} \pm 1^\circ\text{C}$. Glycerin-water solutions are chosen due to the fact that surface tension remains similar throughout, while the viscosity changes more drastically. De-ionized water has a surface tension of 72.8 mN/m and glycerin has a surface tension of 64.0 mN/m . The test fluid with the highest viscosity ($15.2\times$ greater than de-ionized water) is a mixture of 65% glycerin by weight. Its surface tension is therefore 67.1 mN/m or 7.8% lower than that of de-ionized water.¹² However, fluids with significantly lower surface tensions and higher viscosities have been successfully used with the previous design. For low viscosities, it is seen that the range of droplet diameters for a fixed nozzle size remains approximately the same, but the pulse widths required to produce droplets increase. As the viscosity is further increased, the range of droplet sizes does eventually increase.

Overall, the day-to-day reliability of the droplet generator, taken to be one standard deviation over the mean, was found to be 0.9%. This was calculated by repeatedly draining the apparatus and flushing the system with isopropyl alcohol and then filling the system and testing a particular pulse width. The largest drop-to-drop variation was found to be 0.5% over all trials; however, the drop-to-drop variation in a single trial was typically well below 0.2%. Assuming access to a 3D-printer and computer, the complete device can be readily reproduced for less than \$200.

In the [supplementary material](#) associated with this note, we have included a complete parts list, all of the necessary CAD files, a sample Arduino code, and a diagram of the simple circuitry used for the generator, with the aim of making this technology as reproducible and adaptable as possible.

The authors would like to acknowledge the assistance of Olivia Banks with the preparation of the CAD files.

¹D. M. Harris, T. Liu, and J. W. M. Bush, *Exp. Fluids* **56**, 83 (2015).

²A. L. Yarin, *Annu. Rev. Fluid. Mech.* **38**, 159 (2006).

³C. Josserand, P. Ray, and S. Zaleski, *J. Fluid Mech.* **802**, 775 (2016).

⁴W. Zhang, T. Yu, J. Fan, W. Sun, and Z. Cao, *J. Appl. Phys.* **119**, 114901 (2016).

⁵D. Khojasteh, M. Kazerooni, S. Salarian, and R. Kamali, *J. Ind. Eng. Chem.* **42**, 1 (2016).

⁶J. Zou, P. F. Wang, T. R. Zhang, X. Fu, and X. Ruan, *Phys. Fluids* **23**, 044101 (2011).

⁷J. C. Yang, W. Chien, M. King, and W. L. Grosshandler, *Exp. Fluids* **23**, 445 (1997).

⁸D. Terwagne, F. Ludewig, N. Vandewalle, and S. Dorbolo, *Phys. Fluids* **25**, 122101 (2013).

⁹J. Castrejón-Pita, G. Martin, S. Hoath, and I. Hutchings, *Rev. Sci. Instrum.* **79**, 075108 (2008).

¹⁰S. Cheng and S. Chandra, *Exp. Fluids* **34**, 755 (2003).

¹¹J. Canny, *IEEE Trans. Pattern Anal. Mach. Int.* **PAMI-8**, 679 (1986).

¹²E. Muller, Sitzber. Akad. Wiss. Wien, Math.-Naturw. Klasse, Abt. IIa **133**, 133 (1924).

Supplementary Material for
“A versatile 3D-printed droplet-on-demand generator”
Nikolay Ionkin and Daniel Harris

Item	Item Description	Part Number / File Name	Material	Quantity
3D Printed Components				
Fluid Chamber	Used to hold working fluid	Fluid Chamber	(RS-F2-GPCL-04) Resin	1
Fluid Reservoir	Used to hold working fluid	Fluid Reservoir	(RS-F2-GPCL-04) Resin	1
Fluid Reservoir Lid	Used to cover fluid reservoir	Fluid Reservoir Lid	(RS-F2-GPCL-04) Resin	1
Mounting Board	Used to mount fluid chamber and reservoir	Mounting Board	(RS-F2-GPCL-04) Resin	1
Clamping Ring	Used to secure piezoelectric disk to fluid chamber	Clamping Ring	(RS-F2-GPCL-04) Resin	1
Leveling Base	Used to hold all components	Leveling Base	(RS-F2-GPCL-04) Resin	1
Nozzle	Point of delivery for working fluid	Nozzle	(RS-F2-DMBE-02) Resin	1
Tubing Components				
Tubing: 1/8" ID	Used to deliver working fluid to various components	McMaster 5006K63	PVC	4 ft
Tube Fitting: for 1/8" ID tubing	Used to mount tubes to components	McMaster 5454K62	Brass	4
Mounting Hardware				
Bolt: 1/4-20 x 2-1/2"	Secure fluid chamber to mounting board	McMaster 92196A552	18-8 Stainless Steel	2
Bolt: M3 x 14	Secure clamping ring to fluid chamber and fluid reservoir lid to reservoir	McMaster 91292A027	18-8 Stainless Steel	12

Bolt: 1/4-20 x 3-1/2"	Secure fluid reservoir to mounting board	McMaster 92196A556	18-8 Stainless Steel	4
Bolt: 1/4-20 x 1/2"	Secure mounting board to mounting base	McMaster 92196A537	18-8 Stainless Steel	2
Leveling Mount: 1/4-20 Stud	Used to level base	McMaster 23015T82	Plastic w/ Steel Stud	4
Hex Machine Nut: 1/4-20	Used to level base	McMaster 91841A029	18-8 Stainless Steel	4
Flat Washer: 1/4	Used with 1/4-20 bolts	McMaster 92141A029	18-8 Stainless Steel	6
Dowel Pin: 1/8x1/2	Used to hold fluid chamber and reservoir at mounting board	McMaster 97395A441	316 Stainless Steel	4
O-ring: 1/16 Width, Dash No. 026	Seal piezoelectric disk to fluid chamber	McMaster 9452K113	70 Shore A Buna N	1
O-ring: 1/16 Width, Dash No. 014	Seal nozzle to fluid chamber	McMaster 9452K58	70 Shore A Buna N	1
Electrical Components				
Adjustable Power Supply (0-30V)	Used as power supply for piezo	TekPower TP3005N	N/A	1
Arduino Uno Board	Used to control piezoelectric disk via H-bridge	Sparkfun DEV-11021	N/A	1
SN754410 Quad Half H-Bridge	H-bridge integrated chip	Sparkfun COM-00315	N/A	1
Piezoelectric Element: CEB-35D26	Used to force fluid through nozzle	Digikey 102-1128-ND	N/A	1

Arduino Code for “A versatile 3D-printed droplet-on-demand generator”

```
/**
 * Arduino code to control Droplet Generator (Paste Directly into Arduino Desktop App)
 * Nikolay Ionkin & Daniel Harris
 **/

//define the two direction logic pins
const int DIR_A = 5; //pin 5 on Arduino Uno board
const int DIR_B = 4; //pin 4 on Arduino Uno board
const int Trigger = 2; //pin 2 on Arduino Uno board (used to trigger camera if used)
const int DelayTime = 15; //delay between the droplet generation and the camera trigger in ms
-- Typically 15-80 ms (This is not needed if no camera is used)
const int Pulsewidth = 1000 //pulse width for which the piezoelectric actuator is activated in us

void setup(){
  //set all pins as output
  pinMode(DIR_A, OUTPUT);
  pinMode(DIR_B, OUTPUT);
  pinMode(Trigger, OUTPUT);
}

void loop(){

  noInterrupts();
  digitalWrite(DIR_A, HIGH); // Turn on left : high low
  digitalWrite(DIR_B, LOW);
  delayMicroseconds(Pulsewidth); //pulse width for which piezoelectric actuator is activated in
microseconds
  digitalWrite(DIR_A, LOW); // Turn on right : low high
  digitalWrite(DIR_B, HIGH);
  interrupts();

  delay(DelayTime); //delay between the droplet generation and the camera trigger in ms --
Typically 15-80 ms

  digitalWrite(Trigger, HIGH); //triggers pin 2 on Arduino to send signal to camera
  delayMicroseconds(6); //minimum pulse width for Mako U-130B camera is 6 microseconds.
(This is minimum time needed for camera to receive trigger signal)
  digitalWrite(Trigger, LOW); //triggers pin 2 on Arduino to stop sending signal to camera
  //interrupts();

  delay(500); //delay between droplets in ms
}
```

H-Bridge Circuit schematic with Arduino and Camera

